

Credit Portfolio Risk Analysis

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Portfolios: Portfolio 1 and Portfolio 2

Course: Introduction to Credit Risk and Applications in Python

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1. Introduction

This report analyzes the one-year credit risk of two loan portfolios using Monte Carlo simulation. The main goals are to estimate the loss distribution of each portfolio and to quantify the impact of default correlation on tail risk measures such as Value at Risk and Expected Shortfall. Two models are implemented and compared: an Independent Bernoulli baseline and a One-Factor Model.

2. Data

The main analysis uses six raw data files: `entities_info.csv`, `portfolio_1.csv`, `portfolio_2.csv`, `default_history_20y.csv`, `stock_returns.csv`, and `industry_index_returns.csv`. The analysis focuses on Portfolio 1 and Portfolio 2. Portfolio 3 is cleaned in the code but not used in the simulation.

The data sources cover different time periods. The default history contains annual default indicators from 2006 to 2025 and is used to compute historical default rates as a cross-check. The stock return and industry index return files contain daily return observations from 2023 to 2025; these data are used to estimate the factor loading ρ in the One-Factor model. The portfolio files describe the current exposures in 2025.

Before modeling, I standardized entity codes, industry labels, and region labels so that observations could be merged consistently across files. Dates and numeric variables such as EAD, LGD, PD hints, returns, and default indicators were converted to usable formats. Duplicate entity holdings in a portfolio were aggregated by summing EAD, while duplicate default-history records for the same entity and date were resolved by keeping the maximum default indicator.

For the final modeling tables, missing EAD values were replaced with the portfolio median. Missing LGD and PD inputs were filled using industry-level medians, based on the cleaned entity information. I use `base_pd_hint` as the main PD input, while historical PD is kept only as a cross-check because many firms have zero observed defaults in the 20-year default history. Return observations with missing values were removed before estimating factor loadings. The cleaned datasets and summary files are saved in the `results` folder.

3. Methodology

3.1 Model choice

Two models are implemented. The *Independent Bernoulli* model assumes all defaults are independent and serves as a baseline. The *One-Factor Model* introduces correlation through a single common factor. When $\rho = 0$, the systematic factor drops out of the One-Factor model and defaults become independent, so the Independent Bernoulli model is a special case of the One-Factor model with $\rho = 0$. Comparing the two therefore

isolates the pure effect of default correlation on tail risk. Both models use the same PD, LGD, and EAD inputs from the modeling tables constructed in Section 2.

3.2 Model 1: Independent Bernoulli

Each obligor i defaults independently with probability PD_i :

$$Y_i = \mathbf{1}\{U_i < \text{PD}_i\}, \quad U_i \sim \text{Uniform}(0, 1), \quad \text{independent across } i.$$

where $\mathbf{1}\{\cdot\}$ denotes the indicator function.

No additional parameter estimation is required.

3.3 Model 2: One-Factor Model

A latent credit-quality variable is defined for each obligor:

$$X_i = \rho_i F + \sqrt{1 - \rho_i^2} \varepsilon_i, \quad F, \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, 1),$$

where F is a systematic factor shared by all obligors and ε_i is idiosyncratic. Obligor i defaults when X_i falls below the threshold

$$C_i = \Phi^{-1}(\text{PD}_i),$$

which is obtained by inverting the standard normal CDF. This ensures that the unconditional default probability equals PD_i by construction. In the simulation, C_i is computed as `scipy.stats.norm.ppf(PD_i)`.

Conditioning on a realization of the common factor $F = y$, the default events become independent across obligors and the conditional default probability for obligor i is

$$p_i(y) = P(X_i \leq C_i \mid F = y) = \Phi\left(\frac{C_i - \rho_i y}{\sqrt{1 - \rho_i^2}}\right).$$

The common factor F can be interpreted as a macroeconomic state variable. After conditioning on F , defaults are independent Bernoulli variables with factor-dependent default probabilities. When F takes a large negative value, the conditional default probabilities rise at the same time, producing larger joint losses than in the independent model. In the code, I implement the same idea directly by simulating X_i and checking whether it falls below the threshold C_i .

The portfolio loss rate in scenario s is

$$L_s = \frac{\sum_i \text{EAD}_i \cdot \text{LGD}_i \cdot \mathbf{1}\{X_{i,s} \leq C_i\}}{\sum_i \text{EAD}_i}.$$

3.4 Parameter estimation

PD, LGD, EAD. Taken from the modeling tables in Section 2.

Factor loading ρ . The factor loading ρ_i is only required for the One-Factor model. I estimate it by regressing each firm’s daily stock return on the corresponding industry index return:

$$r_{i,t} = \alpha_i + \beta_i \cdot r_{\text{ind},t} + e_{i,t}.$$

The regression provides an empirical measure of how strongly a firm’s return co-moves with its industry index. In a simple one-variable OLS regression, the coefficient of determination satisfies $R^2 = r^2$, where r is the Pearson correlation between the two return series. In the One-Factor model, ρ_i^2 can be interpreted as the share of the latent credit-quality variance explained by the common factor. I therefore use the return-based estimate

$$\rho_i = \sqrt{R_i^2} = |r_i|.$$

This is an empirical proxy rather than a direct observation of asset correlation. As discussed by [Zeng and Zhang \(2001\)](#), equity correlations are not perfect proxies for asset correlations, but they are commonly used when asset returns are not directly observable. This is the case in my dataset, which contains stock returns and industry index returns but no direct asset value process.

In the final cleaned data, all firms in Portfolio 1 and Portfolio 2 have a known industry label, so each firm is matched to its corresponding industry index for the regression. The estimated factor loadings are therefore based on the available overlapping daily stock and industry-index return observations from 2023–2025. The 20-year default history is not used for estimating ρ , because it contains annual default indicators rather than return series. The portfolio-level parameter summary is shown in Table 1.

Table 1: Summary of model inputs by portfolio.

	Portfolio 1	Portfolio 2
Number of obligors	30	30
Total EAD (USD million)	712.7	798.7
Weighted average PD	1.99%	3.53%
Weighted average LGD	45.8%	49.2%
Mean factor loading ρ	0.441	0.545
Implied asset correlation (ρ^2)	19.5%	29.7%

3.5 Simulation setup

The one-year loss distribution is simulated using Monte Carlo with $N = 50,000$ scenarios and a fixed base random seed of 42; each of the four simulations (two portfolios $\times$ two models) uses a distinct seed (42–45) to ensure independence across runs. The following risk measures are computed from the empirical loss distribution.

Expected Loss: $\text{EL} = N^{-1} \sum_s L_s$.

Value at Risk at level α : $\text{VaR}_\alpha = \hat{F}^{-1}(\alpha)$, the empirical α -quantile.

Expected Shortfall at level α : $ES_\alpha = E[L \mid L > VaR_\alpha]$, the mean loss in scenarios strictly exceeding VaR.

I report EL and VaR at 95%, 99%, and 99.9%, and ES at 99%, for both portfolios and both models. ES at 99% is emphasised as it is a coherent risk measure and the standard supervisory reference level.

4. Results and Discussion

4.1 Risk measures

Table 2 reports the simulated risk measures. All values are expressed as a fraction of total EAD.

Table 2: Simulated risk measures (loss / total EAD). $N = 50,000$ scenarios, base seed = 42.

	EL	VaR 95%	VaR 99%	ES 99%	VaR 99.9%
P1 — Independent Bernoulli	0.90%	3.67%	5.53%	6.38%	7.57%
P1 — One-Factor	0.90%	4.27%	7.94%	10.22%	13.21%
P2 — Independent Bernoulli	1.74%	5.37%	7.44%	8.51%	9.95%
P2 — One-Factor	1.73%	7.96%	14.49%	18.98%	24.67%

4.2 Model comparison

The expected losses are nearly identical across both models for the same portfolio. This is expected: since both models use the same PD, LGD, and EAD inputs, the mean of the loss distribution is determined by these inputs alone and is unaffected by the dependence structure.

The difference appears entirely in the tail. For Portfolio 1, the VaR at 99% rises from 5.53% to 7.94% — an increase of about 44% — when moving from the Independent Bernoulli model to the One-Factor model. For Portfolio 2, the gap is larger: VaR 99% increases from 7.44% to 14.49%, a difference of approximately 95%. The effect is even more pronounced at the 99.9% level, where Portfolio 2 reaches 24.67% under the One-Factor model versus 9.95% under independence.

The increase in ES is even larger than the increase in VaR, especially for Portfolio 2. This indicates that default correlation not only shifts the VaR threshold upward but also increases the severity of losses beyond that threshold. Therefore, ES gives useful additional information about tail risk.

Figure 1 illustrates the distributional shift for Portfolio 1. The Independent Bernoulli distribution is narrow and concentrated near the expected loss. The One-Factor distribution has a wider spread and a heavier upper tail, because a bad realization of the common factor F raises the conditional default probabilities of all obligors simultaneously, producing large joint losses that are nearly impossible under independence.

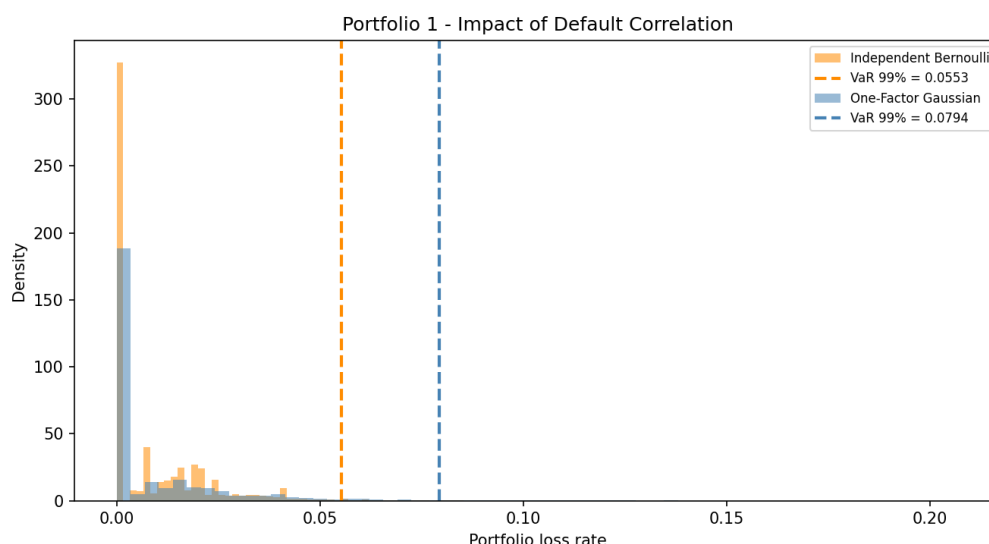


Figure 1: Portfolio 1: loss distributions under the Independent Bernoulli and One-Factor models. Dashed lines mark the VaR at 99%.

4.3 Portfolio comparison

Portfolio 2 is riskier than Portfolio 1 under both models. Under the One-Factor model, its EL is about 1.9 times higher, and its VaR 99.9% is 24.67% compared with 13.21% for Portfolio 1. This is mainly explained by Portfolio 2's higher weighted PD, slightly higher LGD, and higher mean factor loading. Its implied asset correlation of 29.7% is also above the Basel II corporate benchmark range of 12–24% ([Fitch Ratings, 2008](#)).

4.4 Limitations

The analysis has three main limitations. First, factor loadings are estimated from only three years of daily return data (2023–2025), so they may not represent a full credit cycle. Second, PDs and factor loadings are assumed to remain constant over the one-year horizon, although both may change in stress periods. Third, each portfolio contains only 30 obligors, so idiosyncratic risk is not fully diversified away.

5. Conclusion

This report compares an Independent Bernoulli model with a One-Factor model for two credit portfolios. The results show that default correlation has little effect on expected loss but substantially increases VaR and ES, especially for Portfolio 2. This portfolio is riskier because it has higher PD, LGD, and factor loading estimates. Overall, the analysis shows why dependence between defaults is important for portfolio credit risk measurement.

References

- Fitch Ratings. (2008). *Basel II Correlation Values: An Empirical Analysis of EL, UL and the IRB Model*. Credit Market Research, Financial Institutions Special Report.
- Zeng, B., and Zhang, J. (2001). *An Empirical Assessment of Asset Correlation Models*. KMV LLC.